Napari in Practice

From Images to Quantitative Image Analysis Workflows

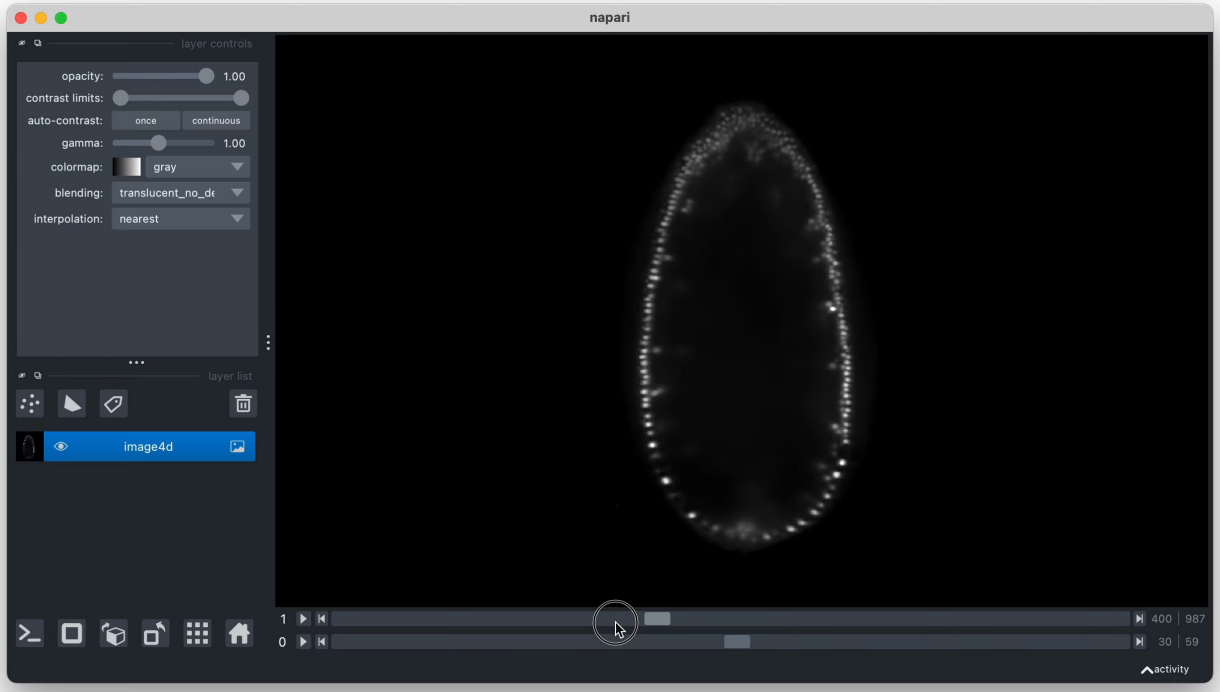
From Click to Code: Reproducibility in Bio-image Analysis

Posted by Mara Lampert, on 5 August 2025

Reproducibility is at the heart of good science. But when it comes to bio-image analysis, it can feel difficult to achieve. Complex datasets, shifting software environments, and countless manual tweaks often make it hard to retrace your own steps and share workflows with others. An aim should be to package the entire analysis, like code, data and environment, in a way that anyone (including future you) could easily run and understand. In this post, we'll explore different ways how and why to save bio-image workflows for reproducibility.

napari: a fast, interactive viewer for multi-dimensional images in Python

- **view and explore** 2D, 3D, and higher-dimensional arrays on a canvas;
- **overlay** derived data such as points, polygons, segmentations, and more;
- **annotate** and **edit** derived datasets, using standard data structures such as NumPy or Zarr arrays, allowing you to
- **seamlessly weave** exploration, computation, and annotation together in imaging data analysis.



Napari Assistant

- The napari-assistant is a napari meta-plugin for building image processing workflows.
- It bridges the gap between manual image interaction and automated Python scripting.

課程大綱

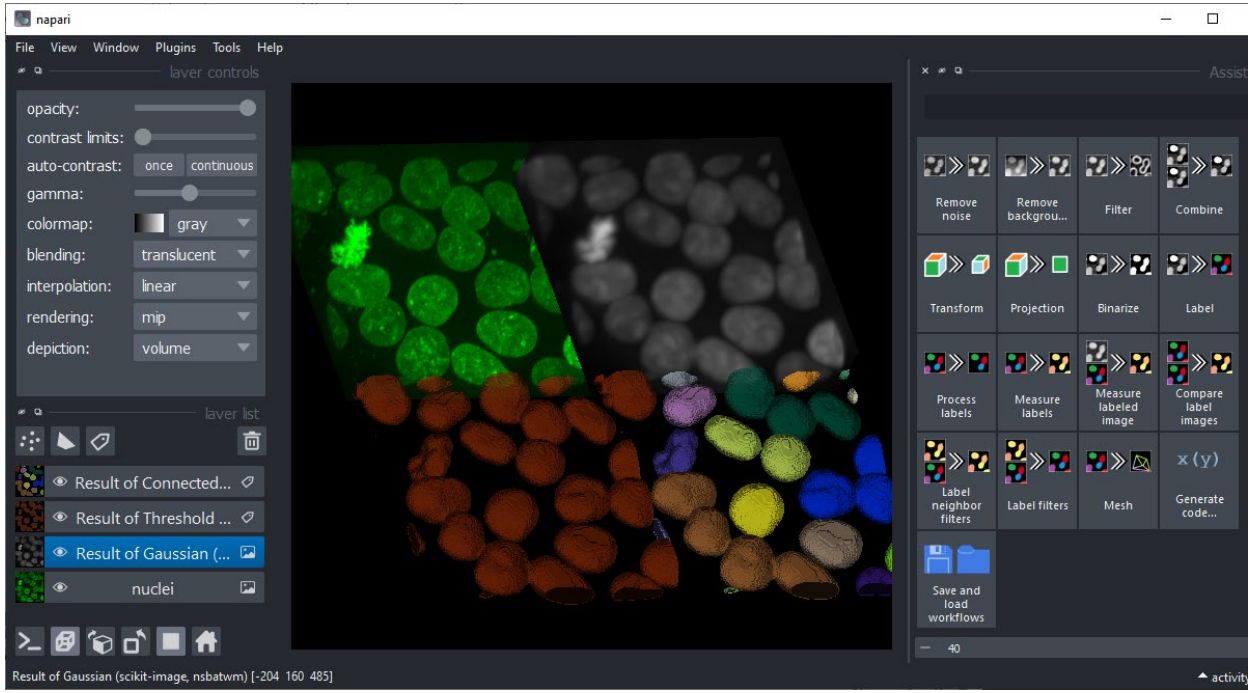
- 1.互動式視覺化基礎 (Interactive Visualization Basics)
  napari GUI 與 assistant 介面與影像分析操作

- 2.程式控制與開發環境整合 (Bridge to Python Code Development)
  建立 Jupyter Notebook 與 napari Viewer 的雙向連動

- 3.建立自動化批次分析工作流 (Batch Processing & Scalable Workflow)
  將單張影像處理轉為可批次的程式化流程

Napari Assistant:

- **Real-time Updates:**
  Changing a parameter in an early step (e.g., Gaussian blur sigma) automatically updates all subsequent layers in the workflow.
- **Workflow Building:**
  Users can chain operations like noise removal, background subtraction, and segmentation by clicking interface buttons.
- **Code Export:**
  Workflows can be exported as **Python scripts** or **Jupyter Notebooks**, allowing users to transition from GUI-based prototyping to reproducible code.



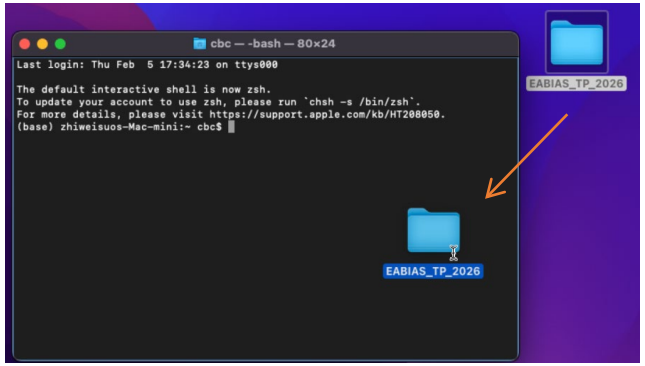
Installation, activation and lauch

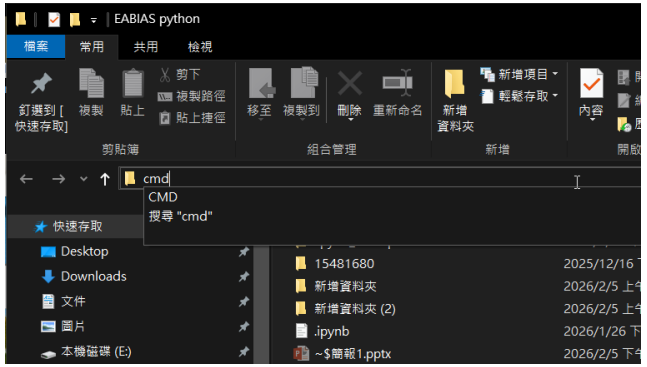
- Installation
  pip install devbio-napari

- Go to 上課資料夾
  Cd c:\user\desktop\eg. EABIAS_TP_2026_Lesson4

- Activate napari environment
  conda activate EABIAS_TP_2026

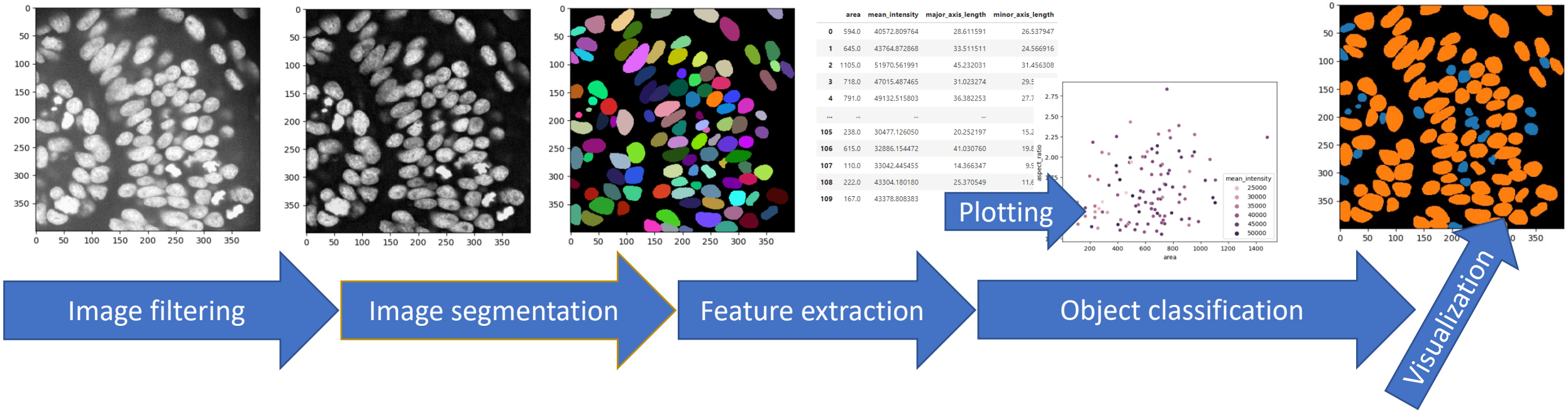
- Start Napari-Assistant
  naparia





Bio-image Analysis Workflows

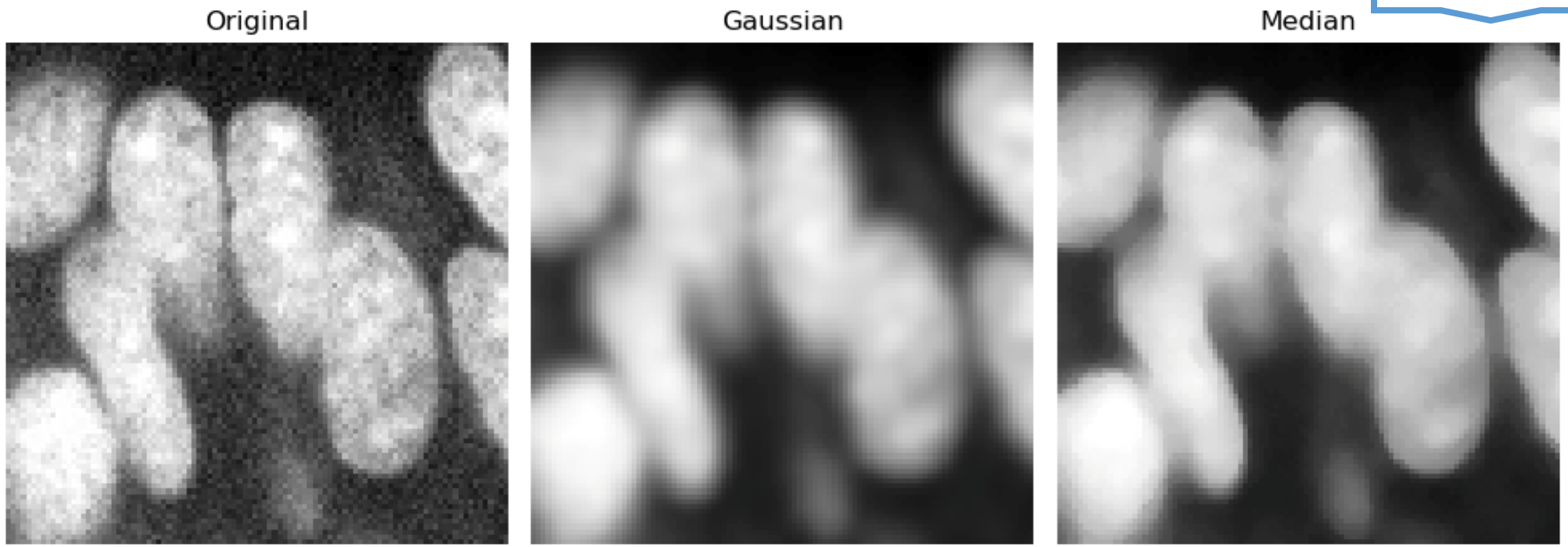
- Goal: **Quantify observations, substantiate conclusions with numbers**



降噪與平滑 (Smoothing / Noise Reduction)

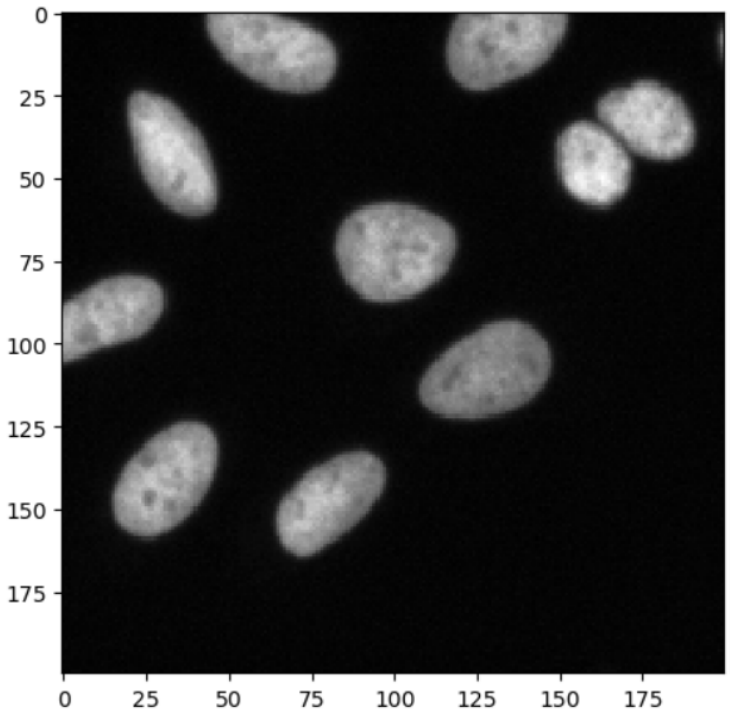
- 高斯濾波器 (Gaussian Blur)
- 中值濾波器 (Median Filter)

“Edge preserving”

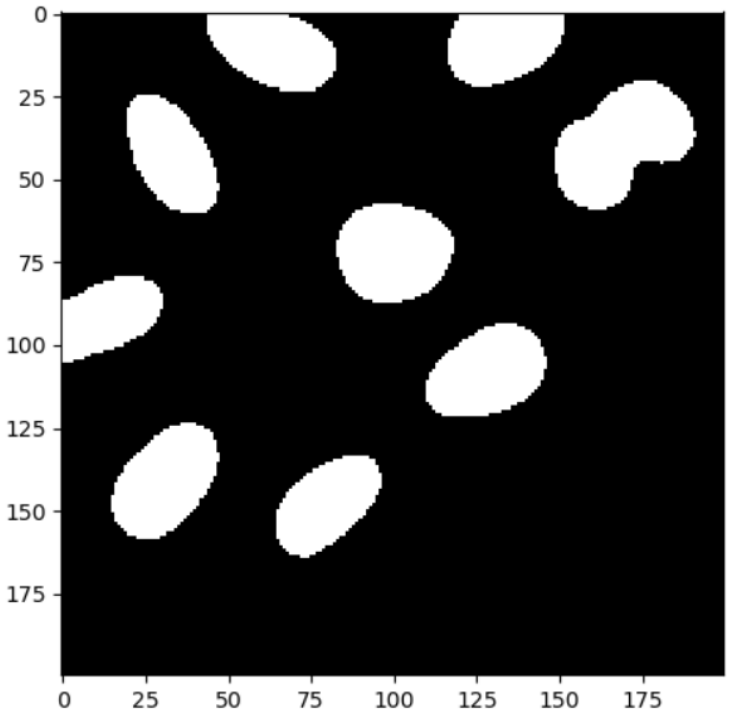


Binarization / Thresholding

Intensity image

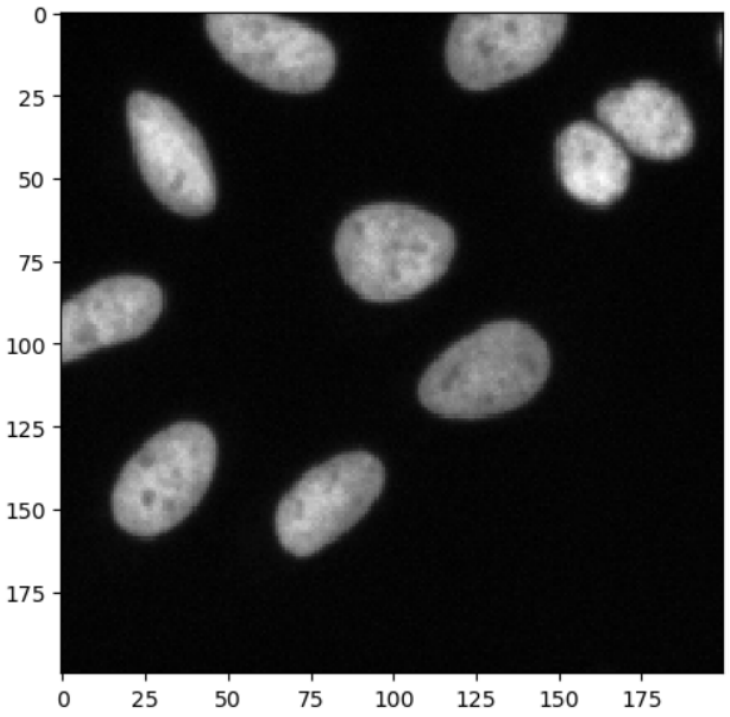


Binary image

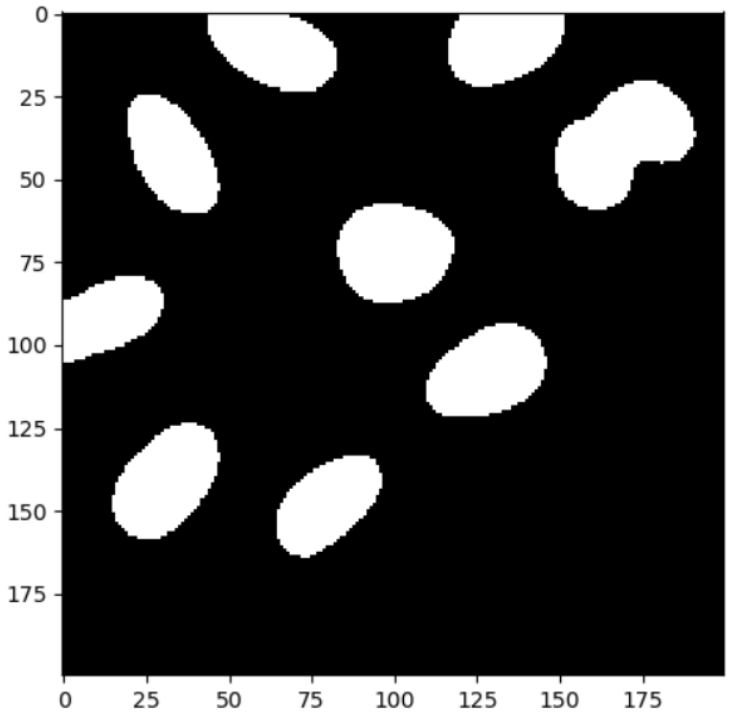


Connected Components Labeling

Intensity image



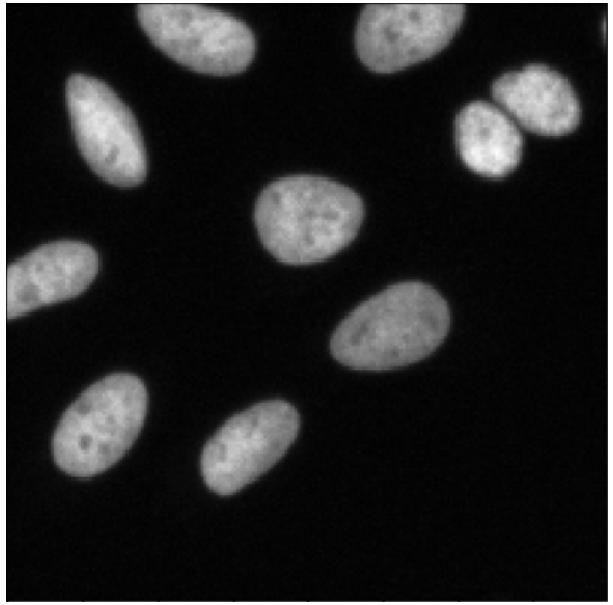
Binary image



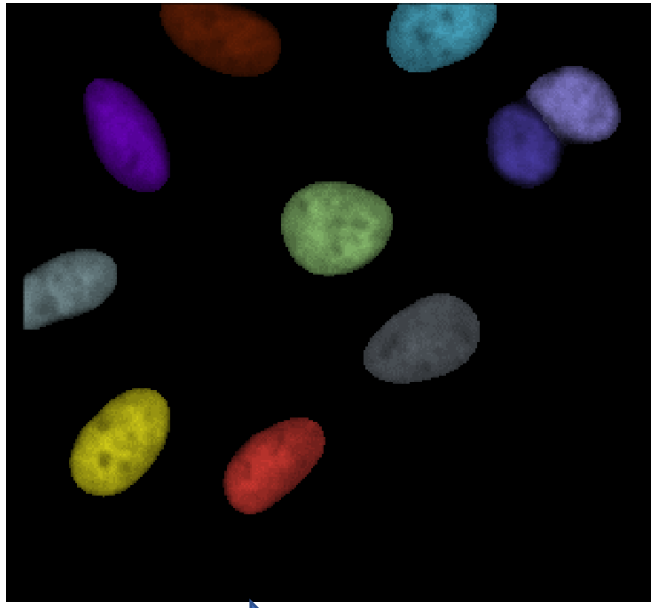
Label image

Measurement

Intensity image

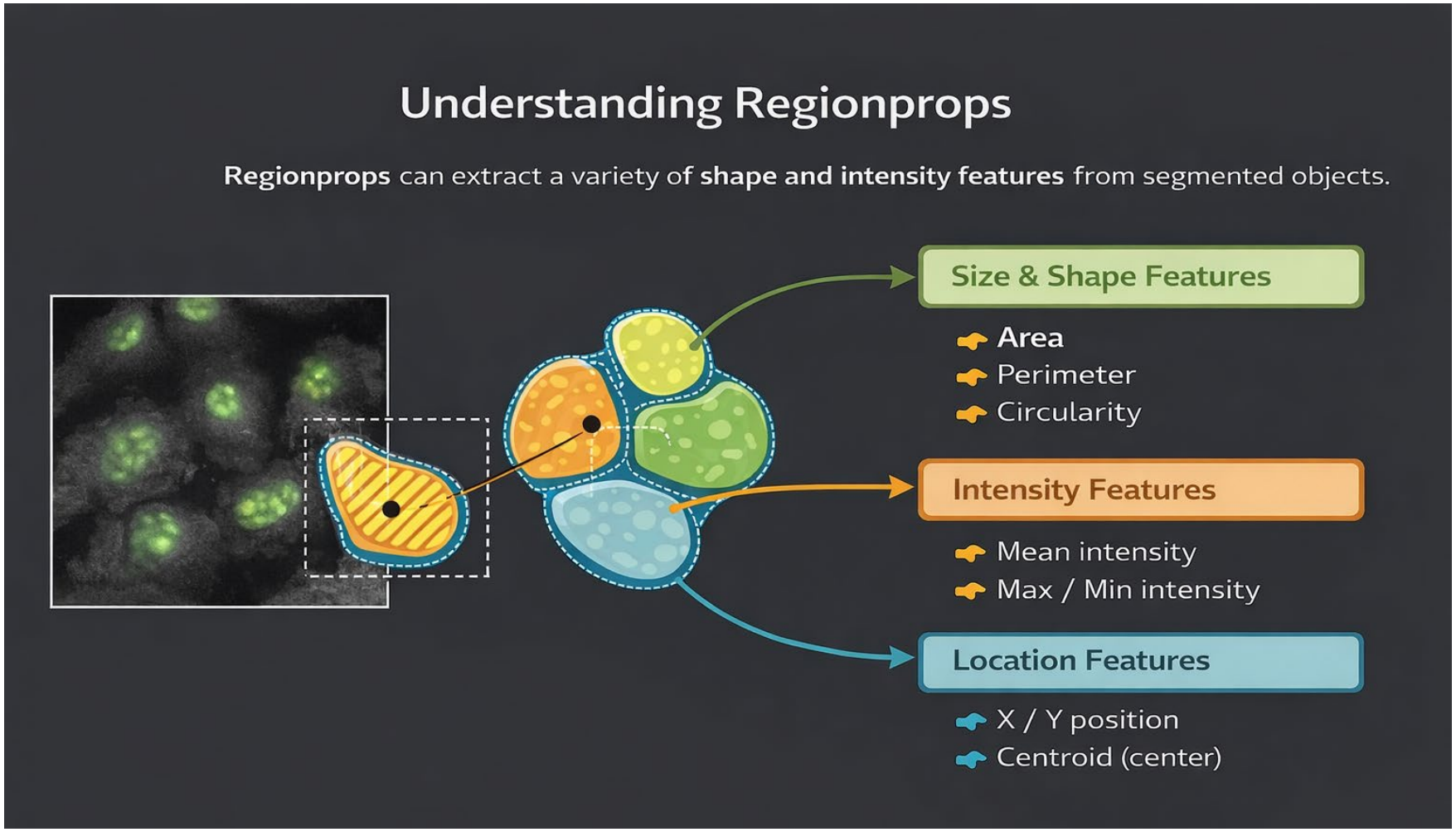


Label image



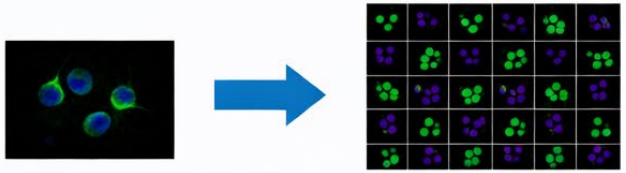
measure

Regionprops
對每個 label（物件）計算特徵（features）



Batch processing

Why Batch Processing?



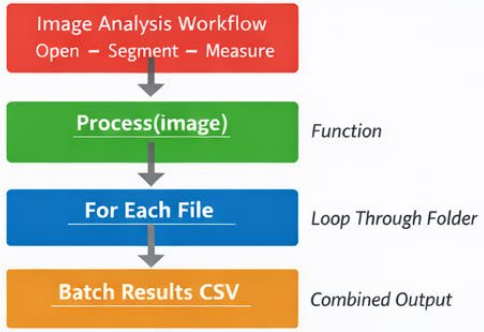
1 Image
Manual Analysis

100 Images
Automated Analysis

Process all images efficiently!

Batch Processing Code

```
for file in input_dir.glob("*.tif"):
    img = imread(file)
    result = process(img)
    all_results.append(result)
```

Read Images → *Run Analysis* → *Save Results*

From Workflow to Batch Processing